ISI 2027 Workshop Proposal: Information Extraction in Practice

Half-day workshop with 20-35 expected participants

Motivation

Information extraction (IE) has transitioned from research prototypes into everyday practice across libraries, archives, and research data services. Large language models (LLMs) have lowered the entry barrier, yet reliable use in production raises open questions about quality assurance, provenance, cost, and sustainability. This half-day workshop brings together researchers and practitioners from information science, digital libraries, and research infrastructures to share concrete experiences with IE in practice. We particularly welcome contributions from the German infrastructure landscape: projects funded by the DFG program "Scientific Library Services and Information Systems" (LIS), the Leibniz information infrastructure institutes, or the National Research Data Infrastructure (NFDI). The workshop aims to identify shared challenges, reusable components, and evaluation practices, and to seed a community of practice beyond ISI 2027.

Workshop Outline

German research infrastructures increasingly rely on IE to make collections and publications machine-actionable. Typical tasks include subject indexing, metadata enrichment, entity linking to authority files such as the GND, and extracting research knowledge from full texts. Much of this work happens in parallel across institutions, with limited exchange about what works in production.

Several lines of activity illustrate this landscape:

- **DFG LIS program:** projects on digitization, specialized information services (Fachinformationsdienste, FID), and information infrastructures regularly include automatic enrichment and extraction components.
- **Leibniz institutes:** for example, ZBW (Kiel/Hamburg) works on automated subject indexing in economics, TIB (Hannover) develops the Open Research Knowledge Graph, GESIS (Mannheim/Cologne) extracts dataset and variable references from social science literature, and FIZ Karlsruhe and ZB MED work on knowledge graphs and life-science literature.
- **NFDI consortia:** communities such as Text+, NFDI4DS, NFDI4Culture, and NFDI4Memory need robust extraction pipelines to connect publications, data, and cultural heritage objects.
- **National library services:** the German National Library (DNB) applies machine-based subject cataloging at scale.

Hamburg, with the ZBW site and the information science programs at HAW in the region, offers a fitting venue. The workshop gives these actors a shared forum within the information science community, which is where questions of evaluation, user needs, and trust in automated metadata have long been studied.

Topics of Interest

We invite presentations and demos on, but not limited to:

- Entity, relation, and event extraction from scholarly, archival, and cultural heritage texts
- LLM-based extraction in production: prompting, fine-tuning, hallucination control, and cost
- Automatic subject indexing and metadata enrichment in libraries
- Entity linking to authority files and knowledge graphs (GND, Wikidata, ORKG)
- Extraction from OCR output, historical documents, tables, and figures
- Evaluation, gold standards, and quality assurance for extracted metadata
- Human-in-the-loop workflows and the role of subject experts and cataloguers
- Provenance, transparency, and FAIR publication of extracted data
- Sustainability: moving IE from funded projects into permanent services
- Legal and ethical aspects, including text and data mining under German copyright law

Format and Schedule

The workshop combines short talks with hands-on and discussion formats, so that participants leave with shared problem statements rather than only a set of presentations. Submissions receive at least two reviews from the program committee.

Time	Session
0:00–0:15	Welcome and introduction
0:15–1:30	Talks and demos from accepted contributions (5 * 10 + 5 min)
1:30–2:00	Coffee Break
2:00–2:45	Breakout groups: evaluation, LLM pipelines, sustainability of services
2:45–3:15	Plenary: group reports and synthesis
3:15–3:30	Wrap-up and next steps

Tentative dates, to be aligned with the ISI 2027 timeline:

- 14.12.2026 Call for contributions
- 25.01.2027 Submission deadline
- 08.02.2027 Notification
- 22.02.2027 Camera-ready for an extended abstract

Target Audience and Expected Outcomes

The workshop addresses information scientists, computational linguists, and data scientists, as well as practitioners in libraries, archives, museums, and research data services. It is equally suited to early-career researchers and to staff responsible for running infrastructure services.

We don't envision a mini-conference and don't focus on publishing proceedings after the workshop. A workshop report where all participants are invited to co-author will be published after the workshop.

Expected outcomes:

- A short joint report on shared challenges and good practices for IE in infrastructures to be published in Datenbank Spektrum (Springer) or another suitable venue.
- A curated, openly available list of tools, datasets, and evaluation resources presented at the workshop
- A starting point for a community of practice, possibly linked to existing NFDI working groups

Organizers

The organizing team combines research and infrastructure perspectives.

- **Philipp Schaer**, TH Köln - Professor for Information Retrieval. Research focus on Information Extraction for retrieval purposes, value-added services for scientific digital libraries, bibliometrics, etc.
- **Konrad Förstner**, ZB MED - Head of Data Science and Services. Professor for Information and Data Literacy.
- **Ralf Krestel**, ZBW & CAU - Professor for Information Profiling and Retrieval. Research on Text Mining, IR, Recommender Systems, LLM and KGs.
- **Benjamin Zapilko**, GESIS - Senior Researcher in the team Information Extraction & Linking. Research on knowledge engineering and knowledge graphs to advance the FAIRness of research infrastructures.

The organizers will evaluate the submission and curate the workshop program.